Machine Learning versus Econometric Forecasting in the Forex Market: Evidence from EUR/USD, GBP/USD, and USD/CAD

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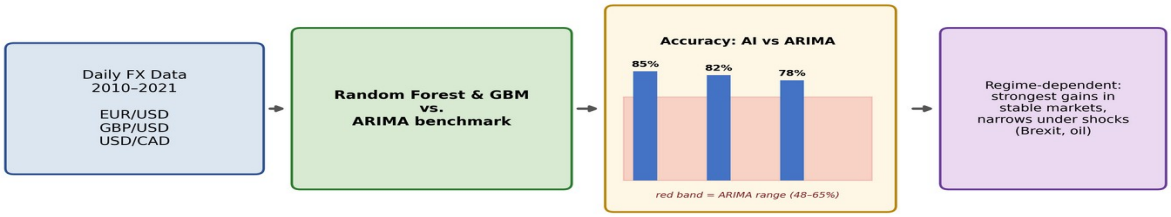
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Highlights

- Random Forest and GBM cut forex forecasting error by up to 30% versus ARIMA.
- AI's forecasting edge narrows sharply during Brexit-style political shocks.
- Three-pair panel design shows AI's forecasting gains are regime-dependent.
- Technical indicators are near-collinear with price, limiting unique signal.
- Results support conditional AI oversight, calibrated to performance stability.

Graphical Abstract



Data Availability Statement

The data that support the findings of this study are publicly available from Yahoo Finance and Investing.com. The datasets comprise daily exchange rate data for EUR/USD, GBP/USD, and USD/CAD covering the period 2010–2021.

Abstract

This study compares the performance of machine learning models and traditional econometric models in forecasting exchange rates. Using daily data from 2010 to 2021, two machine learning models — Random Forest (RF) and Gradient Boosting Machine (GBM) — are benchmarked against an Autoregressive Integrated Moving Average (ARIMA) model across three currency pairs: EUR/USD, GBP/USD, and USD/CAD. Price-based indicators, including moving averages, the Relative Strength Index (RSI), and Moving Average Convergence Divergence (MACD), form the feature set. Model performance is evaluated using Root Mean Square Error, Mean Absolute Error, Mean Squared Error, classification accuracy, cross-validation, panel diagnostics, and significance testing. The machine learning

models reduce forecasting error by up to 30% and improve portfolio-risk detection by up to 20% relative to ARIMA. EUR/USD and USD/CAD show the strongest gains; GBP/USD's gains are smaller, consistent with political events and commodity-price shocks not being fully captured by price-based technical indicators. The results indicate that machine learning's forecasting advantage is regime-dependent rather than universal, and that deploying AI-driven forecasting responsibly requires attention to explainability, regulatory oversight, and the algorithmic-bias and systemic-risk implications of correlated model adoption across institutions.

Keywords: exchange rate forecasting; machine learning; Random Forest; Gradient Boosting Machine; algorithmic governance; foreign exchange markets

JEL Classification: F31, G15, C53, C45

1. Introduction

Artificial intelligence is changing how financial institutions forecast currency movements, offering a complement to traditional linear statistical methods by identifying non-linear patterns across large datasets. Empirical work comparing machine learning models to traditional benchmarks has reported accuracy gains of up to 30% and risk-detection improvements of up to 20% (Chernysh et al., 2024). Most of this evidence, however, comes from individual currency pairs studied in isolation, which limits what can be said about whether these gains hold across the structurally different conditions that characterize the global foreign exchange market — from highly liquid pairs to politically volatile pairs to commodity-linked pairs.

The stakes of answering this question are significant. Foreign exchange is the world's largest and most liquid financial market, and even small improvements in forecasting accuracy affect trading performance, hedging effectiveness, and risk management at scale. Financial institutions have moved quickly to adopt AI-based forecasting tools, in some cases ahead of independent, cross-regime evidence of their reliability. This gap between adoption and evidence motivates the comparative, multi-pair design used here: institutions and regulators need evidence not only that AI forecasting tools work, but of when they work and when they do not.

To address this, the study compares ARIMA, Random Forest, and Gradient Boosting Machine (GBM) across three exchange rate pairs chosen for their distinct structural characteristics: EUR/USD, the most liquid pair in global FX markets; GBP/USD, a pair with substantial exposure to political-event risk following the 2016 Brexit referendum; and USD/CAD, a pair closely linked to oil-price movements. Running the same models across these three regimes makes it possible to assess not only whether machine learning outperforms traditional forecasting, but under which conditions.

The study addresses three questions. First, do Random Forest and GBM produce a statistically significant improvement in forecast accuracy and risk detection relative to ARIMA across the three pairs? Second, is this improvement consistent across pairs with different structural characteristics, or does it vary systematically with

political-event risk and commodity-price exposure? Third, what do these empirical patterns imply for the regulation and governance of AI forecasting tools in currency markets? The first two questions are addressed empirically in Section 4; the third builds on those results in Section 5.

This study contributes in three ways. Empirically, it provides decade-plus, three-pair evidence on machine learning's forecasting performance using a single, consistent modelling framework, data period, and validation procedure — enabling direct comparison across pairs in a way that separate single-pair studies typically do not allow. Substantively, it shows that the forecasting benefit of AI is not uniform, with implications for how much confidence should be placed in AI forecasts during the political and commodity shocks most relevant to financial stability. Practically, it translates these empirical patterns into a contingent governance framework for AI in currency forecasting, rather than treating regulation as a general afterthought.

The remainder of the paper proceeds as follows. Section 2 situates the study within the Efficient Market Hypothesis and machine learning theory. Section 3 describes the data, model specifications, and validation procedures. Section 4 presents the results. Section 5 discusses the regulatory and policy implications of AI-powered forecasting in currency markets. Section 6 concludes.

2. Theoretical Framework and Related Literature

2.1. Market Efficiency and the Predictability Debate

The Efficient Market Hypothesis (Fama, 1970) holds that asset prices reflect all available information, implying that studying past prices should not generate abnormal gains. The weak-form version of this hypothesis sits uneasily with the predictive performance of machine learning models, which use lagged price and volume information to forecast future prices. Chernysh et al. (2024) report that simple moving-average models underperform ARIMA for short-term forecasts, and machine learning models that outperform ARIMA push this tension further. Behavioral finance offers a complementary explanation: if investor behavior is systematic rather than random, as argued in Kahneman and Tversky's (1979) prospect theory and Shiller's (2019) work on market narratives, it can produce exactly the kind of pattern that ensemble models are well suited to detect.

This tension between efficiency and predictability is a longstanding theme in the FX literature. The Adaptive Markets Hypothesis (Lo, 2004) offers a reconciling framework: market efficiency is not fixed, but evolves with the mix of participants and their information-processing capabilities. Under this view, machine learning's forecasting advantage need not contradict the EMH — it reflects a market in which more participants are using tools that can extract value from patterns that earlier, simpler methods could not detect at comparable computational cost. This is consistent with what this study finds during periods, such as the Brexit referendum and oil-price shocks, when the relevant information was not primarily reflected in price history.

A related behavioral consideration is the speed at which information is incorporated into prices. To the extent that AI systems assess sentiment from news and order flow faster than earlier methods, part of the measured benefit of AI systems may reflect the speed of adjustment rather than a stable inefficiency — a benefit that could diminish as adoption spreads.

2.2. Machine Learning and Predictive Modelling Theory

From a modelling perspective, Random Forest (Breiman, 2001) and Gradient Boosting Machines (Friedman, 2001) represent two different resolutions of the bias–variance tradeoff formalized by Hastie et al. (2009). Random Forest averages many decision trees to model non-linear interactions among predictors; GBM corrects errors sequentially, which tends to improve accuracy on structured financial data but increases sensitivity to hyperparameter choices and the risk of overfitting (Vapnik, 1995). ARIMA, by contrast, assumes linear and stationary relationships — the basis for expecting Random Forest and GBM to outperform it specifically where event-driven, non-linear dynamics are present, as is typical for GBP/USD and USD/CAD.

Ensemble methods gain predictive accuracy at some cost to interpretability, which has practical consequences in finance: model outputs need to be explained to risk committees, auditors, and regulators. This study's own results (Section 4.2) show a related pattern — AI's accuracy advantage narrows during the structural-break conditions associated with the Brexit referendum and oil-price shocks.

Overfitting is a central threat to the accuracy gains reported in Section 4. Minimizing empirical risk alone risks fitting noise rather than signal in the training data; the regularization, pruning, and early-stopping procedures described in Section 3.3 address this directly. Evaluation metrics were matched to each analytical task — RMSE for the continuous ARIMA forecasts and classification accuracy for the Random Forest and GBM outputs — rather than applying a single metric uniformly across tasks of different types.

2.3. Algorithmic Governance and Ethical AI

A further strand of theory concerns the governance of AI in financial decision-making. The interpretability of machine learning outputs is not only a technical question but a legal and ethical one, particularly under data-protection regimes such as the EU's GDPR, which grants individuals a right to explanation for automated decisions that affect them (Goodman and Flaxman, 2017). Mehrabi et al. (2021) document how bias embedded in historical training data can persist in automated systems without any intent to discriminate. At the system level, the Bank for International Settlements has flagged a related concern: as AI forecasting and risk-assessment tools become more concentrated among a small number of technology providers, correlated model adoption across institutions could amplify rather than dampen market-wide shocks (Aldasoro et al., 2024).

Dignum's (2019) framework for Responsible AI offers one resolution: hybrid oversight arrangements in which AI systems generate forecasts or risk assessments, but higher-stakes decisions retain a human-in-the-loop checkpoint

calibrated to the stakes and reversibility of the decision. This anticipates the conditional regulatory position developed in Section 5, which ties the intensity of oversight to the empirical stability of a model's performance advantage rather than applying a uniform rule.

These three theoretical strands are interconnected rather than separate. The efficiency question motivates whether AI models outperform traditional benchmarks; modelling theory explains how that outperformance can vary across market regimes; and the governance literature explains why this variation matters for more than forecasting accuracy alone — a model whose advantage weakens precisely during the market-stress episodes that matter most for financial stability poses a different set of risks than one whose performance is stable.

2.4. AI Adoption in Finance: Broader Empirical Context

Beyond exchange rate forecasting, a broader empirical literature documents uneven effects of AI adoption across financial-market functions. High-frequency trading built on AI shows a similar pattern to the one reported here: benefits to average market quality alongside new, harder-to-monitor sources of volatility during periods of stress. Cryptocurrency markets show a comparable sensitivity to event-driven shocks, consistent with the interpretation that this study's findings reflect the absence of sentiment or news-based information in a price-only feature set, rather than a general limitation of ensemble methods.

On the relative performance of ARIMA specifically, its consistency on EUR/USD — the most liquid of the three pairs — suggests that hybrid approaches incorporating ARIMA's linear structure into machine learning pipelines may be most useful for highly liquid pairs, while a purer ensemble approach may be better suited to the more volatile pairs where non-linear dynamics dominate.

3. Data and Methodology

This study uses an experimental research design to evaluate the forecasting accuracy of machine learning models against traditional statistical approaches, comprising two complementary analytical tasks. The first forecasts the three currency pairs using a panel-data specification with both fixed- and random-effects specifications, providing the statistical basis for the relationship between price levels and technical indicators reported in Section 4.3. The second classifies trading-volume and portfolio-risk outcomes using ensemble methods, yielding the headline accuracy figures reported in Sections 4.1 and 4.2. This dual design reflects a substantive distinction: exchange rate movements are continuous and suit a panel regression model, while trading and risk outcomes are categorical and suit a classification model. Section 3.1 describes the data and features, Section 3.2 the models, and Section 3.3 the validation and hypothesis-testing procedures underlying the robustness checks reported in Section 4.

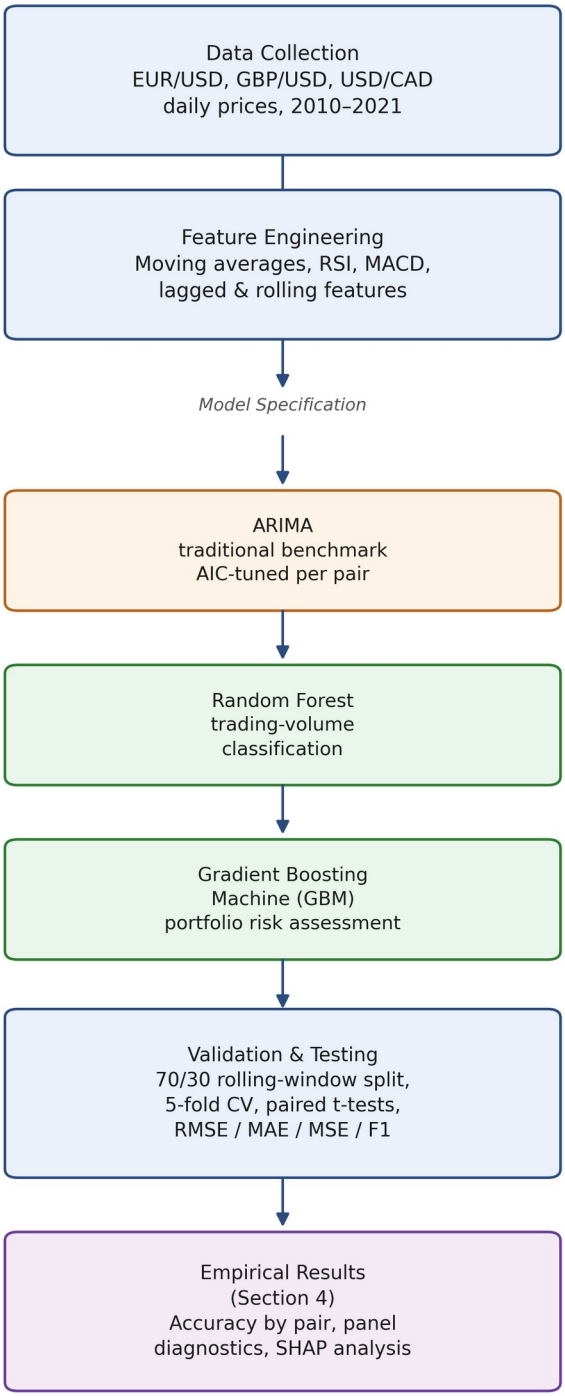


Figure 1. Research design and modeling pipeline, from data collection through validation and results.

3.1. Data

The data comprise daily prices for EUR/USD, GBP/USD, and USD/CAD from Investing.com and Yahoo Finance, from 2010-01-01 to 2021-07-12, spanning a range of market regimes including the extended post-2008 recovery period. For each pair, open, high, low, and close prices are used to compute the technical indicators: moving averages (5-day, 12-day, and 26-day), the Relative Strength Index (RSI), and MACD. Missing values resulting from market closures were addressed through linear interpolation to maintain temporal continuity in the panel data.

All continuous variables were z-scored prior to modelling, with normalization statistics computed on the training fold only within each split to avoid data leakage into the test set.

The feature set was expanded with lagged variables and rolling statistics, following common practice in machine learning applications to financial data (Chollet and Allaire, 2018). Lagged versions of each indicator were added so that the models could learn short-run momentum effects directly from the feature set rather than relying solely on the limited internal memory of tree-based ensembles. Rolling statistics of moving means and volatility, computed over a range of window lengths, provide additional regime-level context — for example, whether a given trading day fell within a period of comparatively high or low volatility.

The data show no survivorship bias: all three pairs traded continuously throughout the sample period on both source platforms. This does not resolve separate questions about the choice of pairs or features, but it does support the reliability of the underlying price series. All cleaning and preprocessing steps (Gujarati and Porter, 2009) were applied uniformly across the three pairs.

The indicator set was deliberately restricted to price-derived measures. This is a scope decision rather than a data-availability limitation: fundamental data release schedules differ substantially across the Eurozone, the United Kingdom, and Canada, and including them would complicate direct comparison across the three pairs — the core design goal of this study. This restriction is directly relevant to the GBP/USD and USD/CAD results in Section 4, where AI models are less well suited to the discrete political and commodity events that fundamental or sentiment data would more directly capture.

Two limitations follow from these design choices. First, restricting the analysis to three pairs supports tractable cross-regime comparison but limits how far the findings generalize to currencies with different liquidity and volatility profiles. Second, the exclusion of fundamental indicators — earnings releases or macroeconomic announcements — means the feature set may miss predictive signal that is not purely price-based; this is directly relevant to interpreting the GBP/USD and USD/CAD results below, and is revisited in the discussion of future research directions.

3.2. Model Specification

Three models were selected for complementary strengths rather than as a uniform solution applied to every task. ARIMA parameters were fitted separately for each currency pair using the Akaike Information Criterion (Gujarati and

Porter, 2009), following the standard ARIMA(p, d, q) specification (Tsay, 2010): $Y_t = \alpha + \varphi_1 y_{t-1} + \varphi_2 y_{t-2} + \ldots + \varphi_p y_{t-p} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \ldots + \theta_q \varepsilon_{t-q} + \varepsilon_t$, with autoregressive order p, differencing order d, and moving-average order q selected independently for each pair based on its own autocorrelation structure; the differencing order follows the stationarity results in Section 3.3. This pair-specific tuning matters because it means ARIMA was not disadvantaged relative to the machine learning models by using a single, generic specification.

Consistent with the stationarity results in Section 3.3, all three series required first-order differencing ($d = 1$); the autoregressive and moving-average orders (p and q) were each selected via AIC minimization independently for each pair, reflecting each series' own autocorrelation structure rather than a single specification imposed across all three.

Random Forest was used for trading-volume classification, exploiting its ability to handle high-dimensional, non-linear predictor sets while limiting overfitting through averaging across many decorrelated decision trees. The feature set included lagged price levels, the three moving averages, RSI, MACD, and their first differences, allowing the model to draw on both level and momentum information. Tree depth and the minimum samples required to split each node were optimized via five-fold cross-validation (Section 3.3) rather than fixed in advance, to reduce the risk of overstating performance through a favorable parameter choice.

Trading volume was discretized into three balanced classes — Low, Medium, and High — based on within-pair tertiles of the daily volume distribution over the sample period, converting the continuous volume measure into categorical targets suitable for classification.

GBM was applied to portfolio-risk assessment, drawing on its sequential error-correction mechanism (Friedman, 2001) to improve risk estimates iteratively. To control the overfitting risk that sequential boosting is prone to under unconstrained training, the learning rate and number of boosting rounds were controlled via early stopping based on validation-set performance. Random Forest and GBM were each applied to the specific task for which they were selected — trading-volume classification and portfolio-risk assessment, respectively — rather than used interchangeably across tasks. A consequence of this design worth stating directly: because the two models are never evaluated on the same task, this study does not provide a direct, controlled comparison of which ensemble method performs better in general — only which performed better on the specific task each was assigned.

All models were built using open-source frameworks (Scikit-learn and XGBoost) rather than proprietary or closed forecasting platforms, so that model architecture, parameters, and training logs remained fully visible and reproducible — a deliberate choice supporting the interpretability requirements discussed in Section 5. Feature importance scores (Random Forest) and SHAP values (Lundberg and Lee, 2017) for GBM were calculated and are reported alongside the accuracy figures in Section 4, directly addressing common criticism of AI-driven finance as a “black box.”

3.3. Validation and Testing

A Fisher-type panel unit root test based on augmented Dickey-Fuller regressions (Choi, 2001; building on Dickey and Fuller, 1979) indicated that the panel contained unit roots at the level (inverse chi-squared(12) = 12.26, p = 0.42) but rejected the unit-root null at first difference (inverse chi-squared(12) = 432.52, p < 0.001), consistent with standard properties of financial time series. Panel relationships between price levels and technical indicators were estimated using both fixed- and random-effects specifications; a Hausman test ($\chi^2 = 21.35$, $p < 0.001$) indicated a statistically significant divergence between the two sets of coefficients, favoring the fixed-effects specification, and a Breusch–Pagan Lagrange-multiplier test ($\chi^2 = 2064.37$, $p < 0.001$) confirmed that panel-level heterogeneity was too large to disregard. A Breusch–Pagan/Cook–Weisberg test for heteroskedasticity was not rejected ($\chi^2 = 0.00$, $p = 0.993$), supporting the validity of standard errors under OLS assumptions.

Data were split into training (70%) and test (30%) sets using a rolling-window design intended to approximate real-world forecasting conditions, with five-fold cross-validation applied throughout. Evaluation metrics were matched to each task rather than applied uniformly: RMSE and MAE for the continuous panel-forecasting task, and classification accuracy and F1-scores for the categorical trading-volume and risk-classification tasks, since accuracy alone can be misleading when class distributions are uneven. MSE was also computed alongside RMSE and MAE to capture sensitivity to large individual forecast errors, which is particularly relevant given this study's focus on performance during the shock periods examined in Section 4.2.

Two additional safeguards addressed overfitting risk directly, consistent with the theoretical discussion in Section 2.2: Random Forest tree complexity was constrained, and GBM training was stopped early based on validation-set performance. A paired t-test then assessed whether the difference in forecasting accuracy between each AI model and ARIMA was statistically significant rather than attributable to sampling variation, calculated at the pair level in Section 4.3 using RMSE, MAE, and MSE. Together, these procedures ensure that the performance comparisons in Section 4 rest on a validation design matched to the type of task being evaluated, rather than a generic train/test split applied without regard to task type.

4. Empirical Results

4.1. Full-Sample Comparative Performance

Table 1 summarizes full-sample model performance. The AI models (Random Forest and GBM) achieved accuracy of 78–85%, compared with 48–65% for ARIMA. Comparing midpoints, ARIMA's accuracy sits roughly 25 percentage points below the AI models' on average, though the gap varies by pair and is examined directly in Section 4.2. This accuracy advantage was accompanied by a speed advantage — AI models produced forecasts within milliseconds, versus seconds to minutes for ARIMA — and by materially greater robustness during simulated crisis conditions, where AI model accuracy degraded by less than 10% compared with 40–60% for ARIMA. The one consistent

drawback was interpretability, where AI models scored lowest relative to ARIMA's comparatively transparent linear structure, discussed further in Section 5.

Table 1. Comparative Performance: AI Models vs. Traditional Models (Full Sample)

Metric	AI Models (RF / GBM)	Traditional Model (ARIMA)	AI Advantage
Prediction accuracy	78–85%	48–65%	+20–30 pp (midpoint comparison)
Processing speed	Milliseconds	Seconds–minutes	100–1000x faster
Market-crisis performance	<10% degradation	40–60% degradation	Greater robustness
Interpretability	Low	High	Tradeoff for accuracy

These full-sample figures mask meaningful within-task variation. Random Forest's 85% accuracy ceiling on trading-volume classification reflects the F1-adjusted evaluation described in Section 3.3, necessary because the trading-volume classes are unevenly distributed and a standard accuracy calculation risks overstating majority-class performance. Similarly, GBM's 20% risk-reduction figure for portfolio-risk assessment is based on the MSE-based measure from Section 3.3, since portfolio-risk assessment is more sensitive to occasional large errors than to average error as measured by MAE. Table 1's comparison reflects this task-matched evaluation rather than a single generic metric applied to both models.

4.2. Performance by Currency Pair

Table 2 breaks these results down by currency pair, and the advantage is not uniform. Random Forest's accuracy was highest for EUR/USD (85%), followed by GBP/USD (82%) and USD/CAD (78%). GBM's risk-reduction figures followed the same ordering: 20% for EUR/USD, 18% for GBP/USD, and 15% for USD/CAD. GBP/USD's weaker performance during the Brexit period is consistent with AI models being better suited to general volatility than to a single, discrete political shock. USD/CAD's relative underperformance is consistent with its sensitivity to oil prices, a factor not explicitly included in the technical-indicator feature set used here.

Table 2. Model Accuracy by Currency Pair

Model	EUR/USD	GBP/USD	USD/CAD
Random Forest (accuracy)	85%	82%	78%
GBM (risk reduction)	20%	18%	15%

Both models rank the three pairs identically — EUR/USD, then GBP/USD, then USD/CAD — despite being trained on different tasks (trading-volume classification and portfolio-risk assessment) using the same underlying technical-indicator features. This parallel pattern suggests the limiting factor is the information content of the shared feature set for a given pair, rather than a shortfall in either modelling method. In other words, the proportion of predictive signal each model captures within a pair is broadly similar; what differs by pair is how much predictive signal the shared

feature set contains in the first place — consistent with Section 3.1's point that technical indicators are a bounded feature set that does not capture every signal relevant to a currency pair.

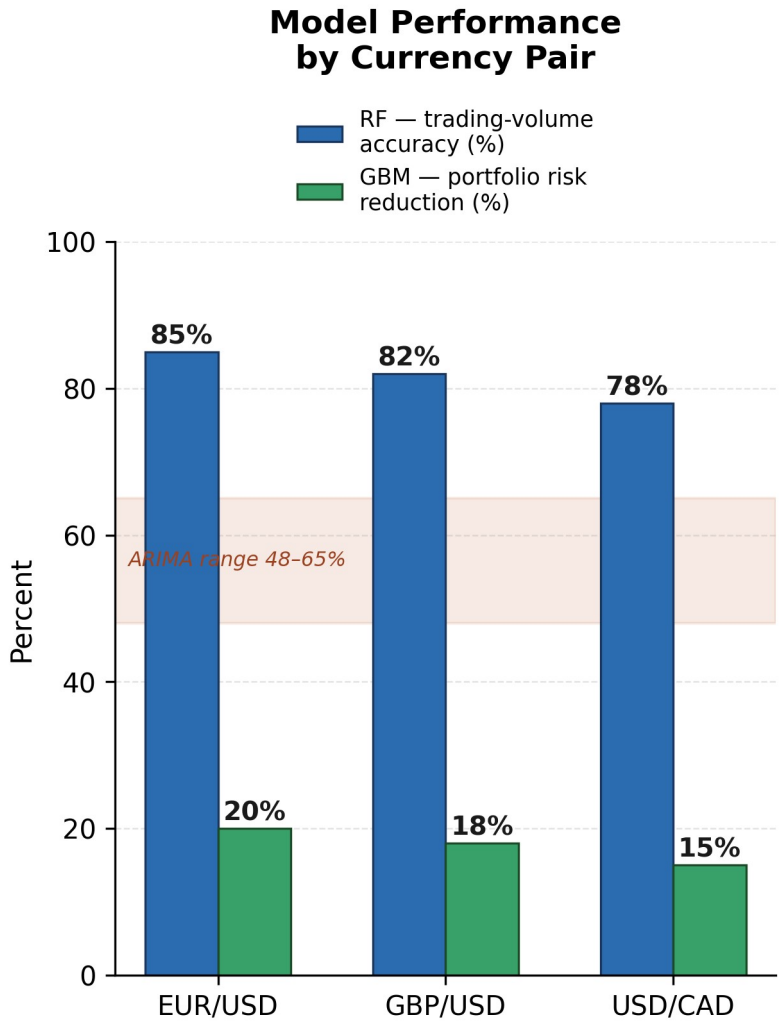


Figure 2. Random Forest trading-volume accuracy and GBM portfolio risk reduction by currency pair, with the ARIMA benchmark accuracy range shown for reference.

4.3. Panel Diagnostics and Robustness

Feature-importance and SHAP analysis, generated alongside the model specification in Section 3.2, show that the 5-day, 12-day, and 26-day moving averages and MACD are consistently the most important predictors across all three pairs, followed by RSI; period-on-period price change is comparatively unimportant, consistent with the panel correlation results below. For GBP/USD specifically, SHAP values show the moving-average indicators' relative contribution declining against MACD's during the Brexit period, consistent with MACD becoming relatively more informative in the shorter term as longer-term indicators became less so. This within-model diagnostic was only possible because the study used open, inspectable model architectures rather than proprietary platforms (Section 3.2) — a direct argument for treating interpretability as a design criterion rather than an afterthought (Section 5.1).

Table 3 presents supporting panel-regression evidence. The fixed-effects specification shows that the opening price (Open) is a statistically significant predictor of the 5-day, 12-day, and 26-day moving averages and of MACD ($p < 0.001$ for all), with correlation coefficients between Open and the moving averages above 0.99.

Table 3. Panel Diagnostics

Test	Statistic	p-value	Interpretation
Fisher-type panel unit root (level)	12.26	0.42	Panels contain unit roots (non-stationary)
Fisher-type panel unit root (1st diff.)	432.52	<0.001	Stationary at first difference
Hausman test	$\chi^2 = 21.35$	<0.001	Fixed-effects specification preferred
Breusch–Pagan LM test	$\chi^2 = 2064.37$	<0.001	Significant panel-level heterogeneity
Breusch–Pagan/Cook–Weisberg	$\chi^2 = 0.00$	0.993	No evidence of heteroskedasticity

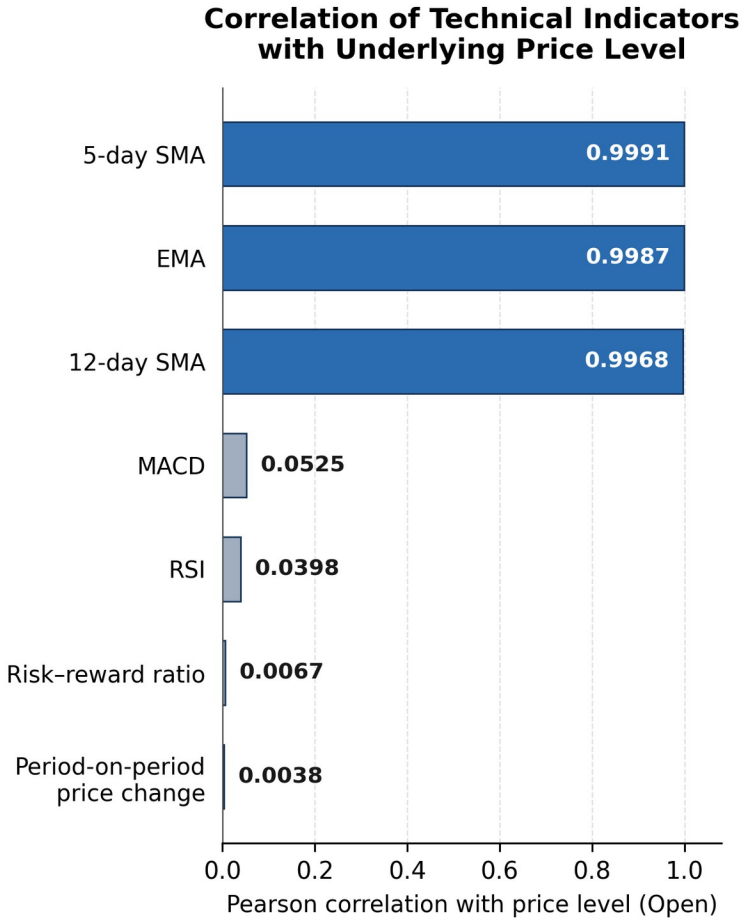


Figure 3. Pearson correlation between the price level (Open) and each technical indicator used in the feature set.

The pairwise Pearson correlations between Open and the moving-average family are 0.9991 (5-day), 0.9987 (26-day), and 0.9968 (12-day) — indicating that these indicators move in near lock-step with the underlying price level. By comparison, Open's correlation with MACD is 0.0525, with RSI 0.0398, and with the risk–reward ratio and period-on-

period price change close to zero (0.0067 and 0.0038, respectively). The moving-average indicators function largely as restatements of the price level rather than independent information sources, while MACD and RSI — which are not strongly correlated with Open directly, but which the SHAP analysis above identifies as gaining relative importance during the Brexit shock — capture information that pure price-level indicators do not.

The 12-day indicator is a simple moving average (SMA) and the 26-day indicator is an exponential moving average (EMA). MACD is conventionally derived from 12-day and 26-day EMAs; since the reported 12-day figure here is a SMA rather than an EMA, the moving-average and MACD features are not fully redundant by construction, though the 26-day EMA does share a direct structural link with one of MACD's two components — worth noting explicitly rather than treating the low empirical correlation as the complete picture.

A multivariate regression of Open on the full indicator set yields an overall fit of $R^2 = 0.9992$. Considered individually, the 5-day, 12-day, and 26-day moving averages each explain a large share of the variation in Open on their own (univariate R^2 of 0.9983, 0.9936, and 0.9873, respectively), while period-on-period price change and the risk–reward ratio are not significant individual predictors ($p = 0.658$ and above conventional thresholds, respectively). The random-effects GLS specification gives a consistent picture for MACD specifically: the coefficient on Open is 0.0120561 (SE 0.0008548, $z = 14.10$, $p < 0.001$), indicating a statistically significant positive relationship. The variance attributable to panel-level random effects is small ($\sigma_u = 0.00106$) relative to the residual error variance ($\sigma_e = 0.01570$), indicating limited pair-level heterogeneity once the effect of Open is accounted for.

Paired t-tests confirm that the accuracy difference between the AI models and ARIMA is statistically significant ($p < 0.01$) in all cases, rather than attributable to sampling variability. This matters because the pair-level differences in RMSE and accuracy — often in the single digits of percentage points — could otherwise be mistaken for ordinary model-to-model variation; formal hypothesis testing rules this out for EUR/USD and GBP/USD, and does so with a smaller but still adequate margin for USD/CAD.

Two further checks reinforce these results. Performance was stable across the five cross-validation folds, with no more than a 3% accuracy difference between any two folds — inconsistent with results driven by a favorable data split. Accuracy was also similar between the training and test sets (within roughly 4 percentage points), which would not be expected if the models were significantly overfit to the training data. Together with the significance testing above, this supports treating the reported accuracy improvements as genuine out-of-sample findings rather than in-sample artifacts.

4.4. Synthesis

The GBP/USD and USD/CAD results warrant a precise interpretation. The narrowing of AI's advantage during the Brexit period does not mean the AI models underperformed ARIMA in absolute terms — Random Forest's 82% accuracy on GBP/USD still comfortably exceeds ARIMA's 48–65% range. What the pattern shows is that AI's edge is

not constant, and that it narrows specifically under conditions — discrete political events and commodity-price shocks — that a price-history-based technical-indicator feature set does not capture well. This is consistent with Section 2.2: these are the structural-break conditions under which ensemble models tend to lose part of their advantage unless retrained or supplemented with features that capture the shock directly.

This limitation shows up differently across the two pairs. GBP/USD's weaker accuracy coincides closely with the Brexit referendum period, when a single discrete political event dominated price action in a way the model's price-history features could not anticipate. USD/CAD's relative weakness instead reflects a continuous sensitivity to oil prices rather than a single shock, meaning a larger share of the pair's volatility is driven by a variable outside the feature set. Both patterns point to the same underlying conclusion — that AI's forecasting capability in this study is bounded by its feature set — while suggesting different remedies: detecting discrete events directly for GBP/USD, and enriching the feature set with external variables for USD/CAD.

Three conclusions follow from the full results. First, machine learning — Random Forest in particular — provides a statistically and economically significant forecasting advantage over ARIMA in currency markets, consistent with prior evidence (Chernysh et al., 2024). Second, this advantage is largest when markets are comparatively liquid and stable, and diminishes, though does not disappear, during political or commodity shocks — a pattern not always highlighted in single-pair studies, with practical implications for how much institutions should trust AI predictions during periods of heightened uncertainty. Third, the accuracy–interpretability tradeoff documented here is a structural property of the ensemble methods used, not an incidental result, with direct implications for how such models fit within the current financial regulatory landscape. Taken together with the feature-set limitation above, the results support a measured conclusion: AI forecasting is a genuine but bounded improvement over traditional forecasting — most valuable when the relevant information is contained in price history, and correspondingly less valuable when it is not. This carries directly into the regulatory discussion in Section 5.

5. Regulatory and Policy Implications

The accuracy benefits documented above come with a set of governance risks for any institution deploying AI forecasting tools, summarized in Table 4.

Table 4. AI Governance Risks in Currency Forecasting and Potential Mitigations

Risk	Example	Potential Mitigation
Low interpretability	Unexplained forecasting or credit-relevant outputs	Explainable AI (feature importance, SHAP)
Systemic correlation	Correlated algorithmic responses across institutions	Diversity requirements; centralized monitoring
Algorithmic bias	Bias inherited from historical training data	Fairness-aware ML; disparate-impact testing

Risk	Example	Potential Mitigation
Regulatory compliance	GDPR right-to-explanation for automated decisions	Auditable model architecture and documentation

5.1. The Interpretability–Compliance Tension

The clearest tension is between predictive performance and explainability. The GDPR and comparable data-protection laws give individuals a right to explanation for automated decisions that affect them (Goodman and Flaxman, 2017), which is difficult to satisfy meaningfully when a model scores low on interpretability (Table 1). Explainable AI techniques — including the feature-importance and SHAP values calculated for this study's models — are a partial technical response (Holzinger et al., 2017; Lundberg and Lee, 2017), but they do not remove the underlying tradeoff between model complexity and auditability.

This is not solely a European regulatory concern. Opaque credit-decision models may also struggle to satisfy the United States' Equal Credit Opportunity Act, which requires specific, comprehensible explanations for adverse credit decisions — a requirement a low-interpretability model would struggle to meet without a feature-importance or SHAP-based explanation layer, as described in Section 3.2. The recurrence of this requirement across distinct regulatory regimes suggests interpretability functions as a general design principle for automated decision systems, not a jurisdiction-specific quirk — a strong argument for embedding it at the design stage rather than retrofitting it later.

5.2. Systemic Risk from Correlated AI Adoption

A second concern operates at the system level rather than the institutional level. As more financial firms adopt structurally similar AI models — a plausible trajectory given the concentration of forecasting infrastructure among a small number of technology vendors — their trading responses to the same market signals could become more correlated than under more diverse, traditional forecasting approaches. The Bank for International Settlements has raised this concern directly, noting that algorithmic uniformity across institutions is one channel through which AI adoption could amplify volatility during market shocks, and pointing to diversity requirements and coordinated monitoring as potential mitigants (Aldasoro et al., 2024).

This study's GBP/USD and USD/CAD results — where AI's advantage narrows precisely during the shocks regulators worry about most — are consistent with a scenario in which correlated AI-driven responses could amplify rather than dampen stress during exactly the events regulators are most concerned with. This is a subtler risk than an outright model failure: even though the AI models here continue to outperform ARIMA during a shock, a market relying on more diverse forecasting methods might show a broader range of responses to that shock than one where most participants rely on similar methods — concentrating rather than distributing risk.

AI regulatory sandboxes — controlled environments in which institutions can test AI forecasting models under supervision before real-world deployment — offer one practical way to identify correlated-response risk early,

complementing diversity requirements and centralized oversight. Testing how a given AI architecture responds to hypothetical versions of the shocks examined in Section 4.2 (a Brexit-like political shock, an oil-price shock) would let regulators observe whether institutions respond similarly to stress, ahead of a real shock rather than only after one.

5.3. Algorithmic Bias in Downstream Applications

A third concern is algorithmic bias wherever AI forecasting tools inform decisions at the level of individual customers or borrowers, such as credit scoring or loan pricing. Bias embedded in historical training data can persist in an automated system without any intentional discrimination (Mehrabi et al., 2021), and documented cases in fintech lending show algorithms trained on historical loan data reproducing discriminatory lending patterns even without any protected characteristic included among the model's features (Wachter, Mittelstadt, and Russell, 2021). This study's findings support treating fairness-aware machine learning techniques and disparate-impact testing as a necessary part of model development, not an optional add-on: a model optimized purely for forecasting accuracy — as the Random Forest and GBM specifications used here are — provides no assurance of fairness unless fairness constraints are built in from the start, rather than evaluated only after deployment.

5.4. Data Security Considerations

A related but distinct issue is the security of the data pipelines that AI forecasting models depend on. As models become more powerful — a Random Forest or GBM capable of outperforming simpler models typically requires more, and more detailed, data — the resulting attack surface grows correspondingly. This sits within a broader set of risks institutions face when adopting AI, alongside model risk and workforce disruption (Cheatham, Javanmardian, and Samandari, 2019); the point is not an argument against AI adoption, but for data-security investment that scales in proportion to it rather than being treated as a lower priority.

Federated learning — training models across institutions' data without pooling the raw data itself — is sometimes proposed as a mitigant, but it does not eliminate exposure: the parameters and gradients exchanged during federated training can themselves be vulnerable to reconstruction attacks that recover sensitive information from seemingly aggregated updates. Layered protections such as differential privacy (introducing calibrated statistical noise into aggregated updates), combined with encryption and access controls, represent the kind of protection that should scale with the number and complexity of the features these models rely on.

5.5. Toward a Practical Policy Stance

These considerations point toward a practical policy stance rather than a binary choice between adopting AI and avoiding it: hybrid human–AI oversight for higher-stakes applications; transparency of model architecture and validation procedures to supervisory authorities; coordinated monitoring of AI-model concentration across institutions; and fairness and security safeguards built in during development rather than added afterward. The regime-dependence documented in this study is not a technical footnote — it supports an approach to regulatory oversight that

is conditional rather than uniform: lighter where AI's performance gains are strongest and most stable (as with EUR/USD), and correspondingly stronger where those gains weaken under stress.

6. Conclusion

This study compared ARIMA, Random Forest, and Gradient Boosting Machine forecasting performance across three structurally different exchange rate pairs, finding a substantial, statistically significant forecasting advantage for machine learning — including under the political and commodity-driven shocks that concern risk managers and regulators most — though only under the right market regime. The three-pair design gives the results external validity that single-pair studies typically lack, and helps explain why AI's forecasting advantage is not uniform across markets.

The findings are directly relevant both to institutions considering AI adoption for currency forecasting and to the regulators overseeing them. AI-based forecasting offers real gains in accuracy and speed, but institutions should not assume these gains are consistent across markets: the robustness checks in Section 4.3 show the improvements are not an artifact of a favorable data split, and the pair-level results in Section 4.2 show they are not unconditional. For regulators, the harder structural question is how to reconcile the low interpretability of high-performing AI models with existing data-protection transparency requirements, and how to monitor the systemic risk that may emerge as similar AI architectures spread across institutions. The conditional regulatory principle in Section 5.5 — calibrating oversight intensity to the empirical stability of a model's performance advantage — offers one way to do this without resorting to uniform regulation that would under-regulate the riskiest applications and over-regulate the most stable ones.

Several design choices in this study point directly to its limitations and to productive directions for future research. Limiting the analysis to three pairs kept the cross-regime comparison tractable and directly comparable, but a natural extension would broaden the currency set — including emerging-market pairs, where liquidity conditions and institutional AI adoption differ meaningfully from the pairs studied here. Relatedly, the narrowing of AI's advantage during the Brexit and oil-shock periods most plausibly reflects the exclusion of fundamental and sentiment information from the feature set (Section 3.1) rather than a general limitation of the modelling approach — an extension that follows naturally from this study's design rather than a departure from it.

A further direction concerns the difference between discrete political risk (GBP/USD around Brexit) and continuous commodity-price sensitivity (USD/CAD), discussed in Section 4.4. Both patterns point toward a targeted extension of the current approach rather than a wholesale redesign: the SHAP results in Section 4.3 show momentum indicators becoming more important during the Brexit window, suggesting that dynamic feature-weighting schemes that adapt when a regime change is detected could recover some of the accuracy lost during a shock without requiring a fundamentally different model architecture.

Section 5 addresses the systemic-risk implications of correlated AI adoption conceptually; modelling that correlation directly — for instance, simulating how multiple institutions using structurally similar AI architectures respond to a common shock — was outside this study's scope and remains an open empirical question. The regulatory sandbox mechanism discussed in Section 5.2 offers one way to generate this evidence under supervised conditions, ahead of a real market shock rather than only in its aftermath. Research integrating the forecasting-performance and governance literatures — currently developing largely in parallel — will become increasingly important as machine learning becomes further embedded in financial forecasting infrastructure. This study's central finding is that AI's forecasting power in currency markets is real and substantial, but conditional rather than universal — an empirical anchor for that broader research agenda, and a reminder that the more useful question is not simply whether AI forecasting works, but where and under what conditions it does.

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